



CV2090

Probability & Statistics for Civil Engineers

(Jul – Nov 2026)

Dr. Prakash S Badal

L16
Aug 27, 2026

Announcement

- 31st Aug: Quiz-1 (8 AM) *NOT part of Quiz-1*
 - Quiz-1 syllabus: up to random variables (including normal, Lognormal, and ~~χ^2~~ , but excluding functions of rv)
 - Check practice problems on GitHub
- Tutorial on Monday (31st Aug)?

- Check course page on GitHub regularly

<https://github.com/prakashvs613/2026-CV2090-IITM/>

Messages from the Last Class

• Message 1: Normal distribution

→ Additive effects → Normal dist
 → PDF, $f_X(x) = \frac{1}{\sqrt{2\pi}\sigma} e^{-\frac{(x-\mu)^2}{2\sigma^2}}$
 $-\infty < x < \infty$

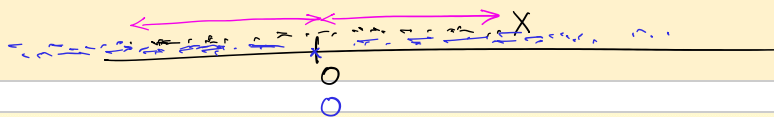
→ Standard Normal RV
 $Z = \frac{X-\mu}{\sigma}$; $\mu_Z = 0$, $\sigma_Z^2 = 1$

→ $X \sim \mathcal{N}(\mu, \sigma^2)$ & $Y = aX + b$
 $\mu_Y = a\mu + b$, $\sigma_Y^2 = a^2\sigma^2$ ①

→ $X_1 \sim \mathcal{N}(\mu_1, \sigma_1^2)$ & $X_2 \sim \mathcal{N}(\mu_2, \sigma_2^2)$ → Independent

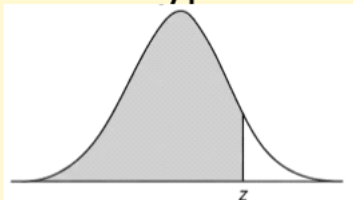
$Y = X_1 + X_2$; $\mu_Y = \mu_1 + \mu_2$; $\sigma_Y^2 = \sigma_1^2 + \sigma_2^2$ ②

→ Let $X_2 = X_1$, $\Rightarrow Y = 2X_1$, $\mu_Y = 2\mu_1$, $(\sigma_Y^2 = 4\sigma^2)$

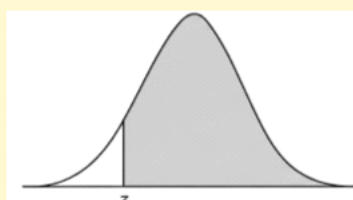
 $\text{Var}[X] = E[(X-\mu)^2]$

Normal distribution

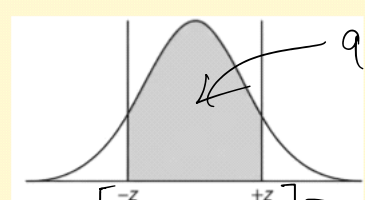
• Three types of areas under the curve are useful



CDF



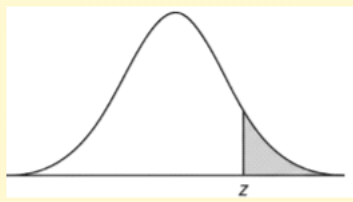
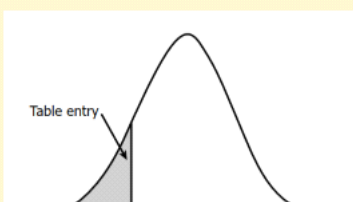
1-CDF

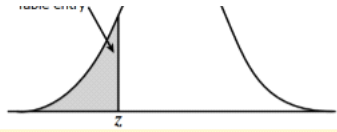


Confidence Interval (CI)

95% CI

$[\mu - 2\sigma, \mu + 2\sigma]$





Lognormal Distribution

Ang and Tang (Section 3.2.2)

Lognormal Distribution: motivation

* Additive effects often lead to Normal Dist.

$$X = X_1 + X_2 + \dots + X_n \sim N(\mu, \sigma^2)$$

* what dist. will "Multiplicative effects" lead to?

$$X = X_1 X_2 \dots X_n$$

e.g., wealth growth, population growth, fatigue loss

$$Y = \ln X = \ln X_1 + \ln X_2 + \dots + \ln X_n$$

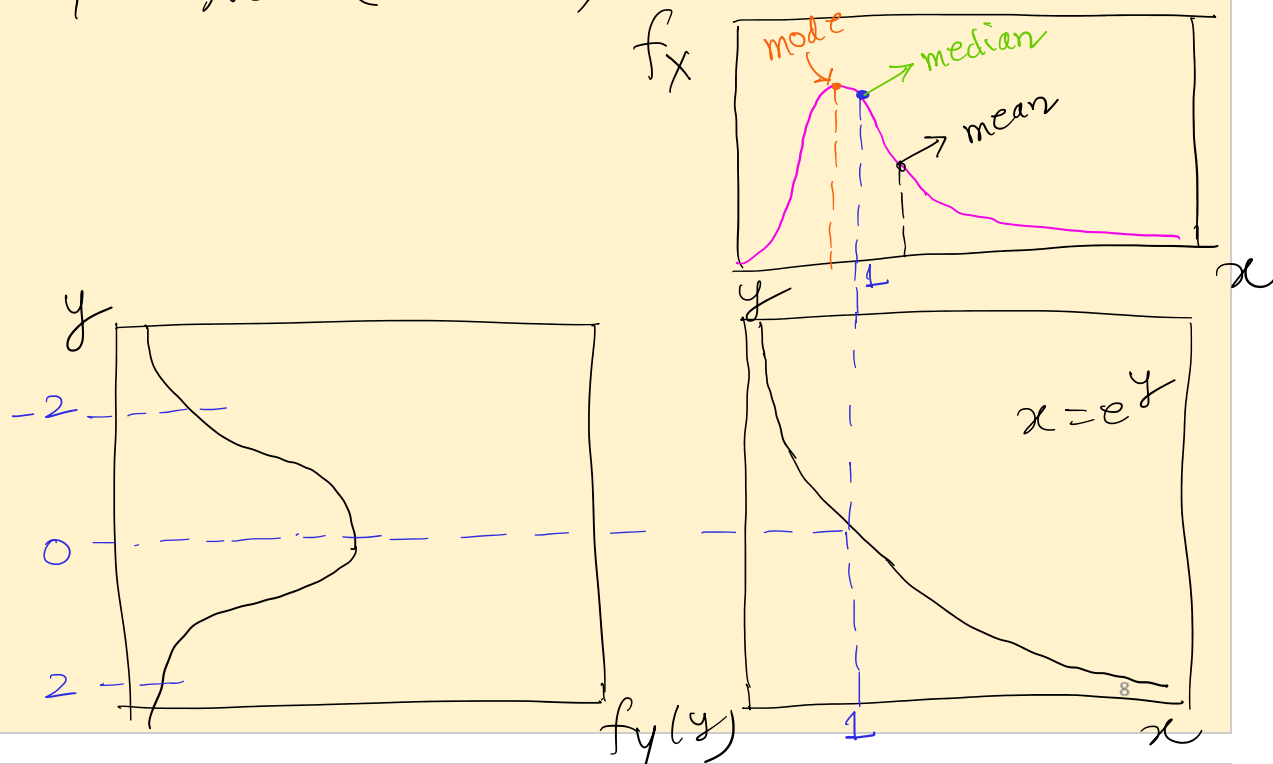
If $\ln X \sim N \Rightarrow X \sim LN$ (lognormal)

Additive effects $\rightarrow$ Normal Dist
Multiplicative effects $\rightarrow$ Lognormal Dist.

Lognormal Distribution: transformation

$$X \sim \text{Lognormal} \iff Y = \ln X \sim \text{Normal}$$

$$Y \sim \text{Normal} \iff X = e^Y \sim \text{Lognormal}$$



Lognormal Distribution: parameters

- A random variable is Lognormal, if its log follows Normal

$$X \sim \text{Lognormal} \iff Y = \ln(X) \sim \text{Normal}$$

Knowing $Y \sim \mathcal{N}(\mu_Y, \sigma_Y^2)$, determine PDF of X

Recall, $f_Y(y; \mu_Y, \sigma_Y^2) = \frac{1}{\sqrt{2\pi} \sigma_Y} \exp \left[-\frac{(y - \mu_Y)^2}{2\sigma_Y^2} \right]$

Jacobian $f_X dx = f_Y dy$

$$\frac{dy}{dx} = \frac{1}{x} \quad \& \quad f_X = f_Y(y) \frac{dy}{dx}$$

$$= f_Y(\ln x) \frac{dy}{dx}$$

$$f_X(x) = \frac{1}{\sqrt{2\pi} \sigma_Y x} \exp \left[-\frac{(\ln x - \mu_Y)^2}{2\sigma_Y^2} \right]$$

μ, σ^2
c.o.v. coefficient of variation,
 $\delta = \sigma/\mu$

$Y \sim \text{Normal}, \mu_Y, \sigma_Y^2$
 $\parallel$
 λ, ζ^2
 $\ln X$

$X \sim \text{Lognormal} \mu_X, \sigma_X^2 \Rightarrow \delta_X = \frac{\sigma_X}{\mu_X}$

$\mu_X = \exp \left(\lambda + \frac{1}{2} \zeta^2 \right)$

$\zeta^2 = \exp(\zeta^2) - 1$

$\ln X$

$$\delta_x^2 = \exp(\zeta^2) - 1$$

Lognormal Distribution

